Normalization Without Convergence: Large Language Models, Linguistic Entropy, and the Myth of Cognitive Homogenization

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Abstract

Recent commentary in cognitive science and computational linguistics has proposed that large language models (LLMs) are driving a broad homogenization of human language, perspective, and reasoning. According to this view, the widespread use of generative systems encourages convergence toward a statistical "median" embedded in the models' training distributions, potentially narrowing the diversity of linguistic expression and epistemic viewpoints across communicating populations. While this hypothesis raises important questions about the long-run cultural consequences of AI-mediated communication, many existing formulations rest on an implicit and under-examined assumption: that linguistic output provides a transparent window into the underlying generative structure of human cognition.

This paper argues that such claims systematically conflate two distinct phenomena: *surface-level normalization in the communication channel* and *deep convergence in the generative processes of human cognition*. Drawing on an information-theoretic framework, we demonstrate that a system capable of mapping many idiosyncratic inputs to a standardized representation necessarily reduces the entropy of observable outputs while leaving the entropy of the underlying source distribution unchanged or unaffected. Formally, this follows from the data-processing inequality: post-processing cannot increase mutual information. In other words, uniformity at the level of representational output does not entail—and cannot establish—homogenization at the level of thought.

From this perspective, large language models function less as normative enforcement mechanisms than as *translation layers* between heterogeneous private linguistic systems and shared public registers. By reducing the communicative cost of nonstandard spelling, dialect, or multilingual composition, such systems may in fact expand the effective expressive space available to individual authors. We formalize this argument through a source–channel

model, develop the notion of *linguistic attractor basins* in a probabilistic landscape, and introduce a distinction between the entropy of generative sources and the entropy of exploratory discovery processes.

We further argue that the more significant ecological risk of LLM-mediated communication lies not in stylistic convergence but in the potential contraction of *exploratory variance* in knowledge discovery: the diversity of trajectories through which individuals encounter novel ideas, conflicting interpretations, and unexpected conceptual recombinations. If model-mediated search increasingly directs users toward the same conceptual summaries and canonical explanations, the space of intellectual trajectories may contract even while expressive diversity at the surface level remains intact. Understanding the cultural consequences of generative AI therefore requires attention not only to the entropy of linguistic form but also to the information ecology of exploration, discovery, and conceptual variation.

Contents

1 Introduction

Technologies that mediate human communication have repeatedly and profoundly reshaped the structure of linguistic expression and intellectual exchange. Writing systems, printing presses, editorial institutions, telegraph networks, search engines, and digital archives have each altered the mechanisms through which language is produced, transmitted, and interpreted. These transformations have recurrently generated concerns about the potential homogenization of thought: that the machinery of communication will ultimately flatten the diversity of the minds it connects. Yet the historical record consistently resists this conclusion. The diffusion of moveable type standardized orthographic conventions across European languages without destroying the intellectual heterogeneity of the Reformation. Editorial institutions imposed consistent rhetorical styles on scientific writing without constraining the diversity of scientific ideas. The internet concentrated written discourse around a small number of dominant platforms without eliminating the extraordinary proliferation of viewpoints that those platforms carry. In each case, normalization in the communication channel coexisted with the persistence or even expansion of cognitive diversity at the level of the source.

The emergence of large language models (LLMs) represents the latest development in this long sequence of communicative technologies. Generative AI systems are now widely used to assist with writing, translation, summarization, coding, and information retrieval. Because these systems produce text by sampling from probability distributions learned from large corpora, their outputs often reflect statistically common linguistic forms: frequently recurring syntactic constructions, standard rhetorical organization, and stylistically moderate registers. This characteristic has prompted a growing body of scholarly commentary suggesting that the widespread adoption of LLMs may gradually homogenize human language, perspective, and reasoning—that individual discourse will be pulled, through repeated interaction with shared systems, toward a common statistical center.

The homogenization hypothesis raises important questions regarding the long-term cultural consequences of AI-mediated communication. If a large and growing proportion of human discourse becomes filtered through a small set of generative systems, the observable distribution of linguistic expression may indeed shift toward a common mode. Such a shift could plausibly influence not only the stylistic surface of language but the structure of argumentation and the framing of ideas. These are legitimate concerns that deserve serious analytical treatment.

However, evaluating the homogenization hypothesis requires careful attention to a distinction that much of the existing literature has failed to draw clearly: the distinction between *observable representations* and the *generative processes* that produce them. Communication systems routinely include intermediate transformation layers that reshape linguistic signals during transmission without altering the cognitive diversity of the communicating agents. Spelling conventions normalize orthographic variation. Editorial practices standardize rhetorical structure. Translation systems project expressions across language boundaries. Each of these mechanisms compresses surface-level variation without penetrating the generative layer of cognition.

Large language models can be interpreted as a technologically novel form of such normalization. When used as editing or translation tools, they map diverse linguistic inputs into standardized outputs. The resulting reduction in observable variability may therefore reflect the operation of the communication channel rather than any transformation of the generative source. If this interpretation is correct, the inference from observed output uniformity to cognitive convergence is formally invalid.

This paper develops a theoretical framework for analyzing this distinction with precision, drawing on concepts from information theory, probabilistic modeling, and the ecology of knowledge systems. The central claim is that the homogenization thesis systematically conflates two distinct kinds of entropy reduction: compression within the communication channel, which is a property of the transformation, and convergence within the generative processes of human cognition, which is a claim about the source distribution.

The analysis is organized as follows. Section 2 introduces the homogenization hypothesis in its principal variants and identifies the shared inferential assumptions that underlie them. Section 3 develops an information-theoretic framework distinguishing source entropy from channel entropy and establishes the central formal results. Section 4 elaborates the distinction between private generative systems and public communicative registers, drawing on historical precedent. Section 5 analyzes the difference between instrumental and epistemic uses of generative models and their differential implications for cognitive influence. Sections 6–7 develop the geometric notion of linguistic attractor basins and the conceptual landscape in which they reside. Section 8 introduces the concept of exploratory entropy and analyzes how algorithmic mediation reshapes discovery distributions. Sections 11 and 12 engage with the specific claims of recent empirical literature and propose a more precise research program. Section 13 situates LLMs within a broader history of cognitive prosthetic technologies. Section 15 draws the threads together and states the paper's principal conclusions. Appendices provide more detailed derivations of the central entropy results and a formal treatment of rank-mediated discovery distributions.

2 The Homogenization Thesis

Recent discussions in cognitive science and computational linguistics have articulated a growing concern that large language models may induce a progressive homogenization of human language, reasoning, and perspective. The core intuition motivating this family of claims is straightforward. Because LLMs are trained on large corpora that aggregate statistical regularities across vast quantities of human communication, their outputs tend to reflect the most probable linguistic forms and conceptual framings present in those datasets. When such systems are widely adopted as writing assistants, editors, and conversational partners, it is argued that human discourse may gradually converge toward the same statistical center that those systems implicitly represent [Sourati et al., 2026].

In this view, the widespread use of generative language models acts as a powerful attractor within the space of linguistic and epistemic practices. Individual expressions that initially vary across dialects, registers, educational backgrounds, and cultural contexts are progressively "pulled" toward a shared normative distribution. Over time, this process is thought to reduce variance across several dimensions of human communication simultaneously: lexical choice, rhetorical structure, argumentative form, and epistemic framing.

The homogenization thesis appears in three structurally distinct but related variants that are worth distinguishing precisely.

2.1 Linguistic Convergence

The *linguistic convergence* hypothesis proposes that texts produced with the assistance of LLMs increasingly resemble one another in vocabulary, syntax, and discourse structure. Because the models generate outputs according to learned probability distributions over sequences of tokens, repeated use of the same systems may produce recognizable stylistic regularities across otherwise independent authors. The distributional signature of the model bleeds into the distributional signature of the assisted writer.

This form of the hypothesis is the most empirically tractable, since it concerns surface properties of text that can in principle be measured directly. Its weakness, as argued below, is that surface similarity among assisted texts is entirely compatible with high diversity in the underlying cognitive processes that generated the pre-normalized inputs.

2.2 Perspectival Convergence

The *perspectival convergence* hypothesis suggests that LLMs encode the dominant assumptions and interpretive frameworks of their training data and transmit those frameworks to users through the authority that attaches to model-generated outputs. Since large portions of the digital corpus from which these models are trained originate in Western academic, journalistic, and institutional contexts, critics argue that the models may implicitly reinforce the epistemic norms, evaluative standards, and categorical schemes associated with those contexts [Bender et al., 2021]. Users who repeatedly encounter these norms may internalize them as default frameworks for understanding the world.

This variant is more difficult to operationalize empirically, since it requires measuring shifts in epistemic orientation rather than surface linguistic properties. It also makes stronger assumptions about the mechanism of influence, implying that repeated exposure to model outputs reshapes the prior beliefs and conceptual categories of users in measurable ways.

2.3 Reasoning Convergence

The *reasoning convergence* hypothesis proposes that repeated interaction with model-generated explanations, arguments, and structured reasoning traces may influence the inferential strategies that humans employ when solving problems. The structured step-by-step reasoning formats often produced by contemporary language models might, on this view, serve as templates for human inference—gradually reshaping the architecture of human problem-solving toward the heuristics that models have been trained or tuned to produce.

This is the strongest form of the hypothesis and also the least supported by available evidence. Demonstrating reasoning convergence would require longitudinal evidence of durable transfer from model-mediated reasoning contexts to independent problem-solving behavior—a much higher empirical bar than demonstrating stylistic similarity in assisted texts.

2.4 The Shared Inferential Structure

Despite their differences, all three variants of the homogenization thesis share a common inferential structure. Each proceeds from an observation about the properties of LLM outputs or LLM-assisted human outputs to a claim about the underlying generative processes of human cognition. They assume, explicitly or implicitly, that observable linguistic behavior provides a reliable proxy for the internal structure of human thought.

This assumption is the central target of the present analysis. The argument that follows does not deny that generative AI systems influence human behavior—they surely do, as do all tools that mediate communication and cognition. The argument is rather that the specific inference from output uniformity to generative convergence is formally invalid, because a normalization layer in the communication channel can produce the former without the latter.

3 Source Entropy and Channel Normalization

A rigorous analysis of the homogenization hypothesis requires placing the communication process within an information-theoretic framework. In classical information theory, a communication system involves at minimum three components: a source that generates signals, a channel through which those signals are transmitted, and a receiver that interprets the resulting representation [Shannon, 1948]. The statistical properties of what the receiver observes depend on the joint operation of the source and the channel. Crucially, these two contributions need not mirror one another.

3.1 The Communication Model

In the context of LLM-mediated communication, the components of the standard model admit natural interpretations. The *generative source* corresponds to the cognitive and linguistic processes through which a human author formulates and expresses ideas. The *normalization layer* corresponds to the large language model when it is used to edit, rewrite, translate, or otherwise transform the human-generated text. The *communication channel* corresponds to the medium through which the normalized representation is transmitted. The *receiver* corresponds to the audience that interprets the resulting output.

Let X denote a random variable representing expressions drawn from the generative source distribution $p(x)$. Let T denote the deterministic transformation applied by the normalization layer, so that the observable output is $Y = T(X)$. The distribution of Y is given by

$$p(y) = \sum_{\{x:T(x)=y\}} p(x).$$

This is the pushforward of the source distribution under the map T.

3.2 Entropy of the Generative Source

The entropy of the generative source distribution quantifies the diversity of linguistic expressions produced by the human author:

$$H(X) = -\sum_{x} p(x) \log p(x).$$

High values of $H(X)$ indicate that the author draws from a wide and roughly uniform distribution over possible expressions; low values indicate that the author's choices are heavily concentrated around a small set of forms.

3.3 Entropy of the Normalized Representation

The entropy of the observable representation is

$$H(Y) = -\sum_y p(y) \log p(y).$$

When the transformation T is many-to-one—that is, when multiple distinct inputs $x_1, x_2, \ldots$ all map to the same output y—the output distribution $p(y)$ is strictly more concentrated than the source distribution $p(x)$. In this case $H(Y) < H(X)$.

Definition 3.1 (Compression Ratio)**.** The *compression ratio* of a normalization transformation T at a point y in the output space is

$$\kappa(y) = |\{x : T(x) = y\}|.$$

If $\kappa(y) > 1$ for any y, the transformation is non-injective and compresses the source distribution.

Proposition 3.2 (Entropy Inequality)**.** *For any deterministic mapping $T : \mathcal{X} \to \mathcal{Y}$,*

$$H(Y) \leq H(X),$$

with equality if and only if T is injective on the support of $p(x)$.

Proof. Because $Y = T(X)$ is a deterministic function of X, knowledge of X uniquely determines Y. Therefore the conditional entropy satisfies

$$H(Y \mid X) = 0.$$

By the chain rule of entropy,

$$H(X, Y) = H(X) + H(Y \mid X) = H(X).$$

But the chain rule also gives

$$H(X, Y) = H(Y) + H(X \mid Y).$$

Combining these two expressions,

$$H(X) = H(Y) + H(X \mid Y).$$

Since conditional entropy is always nonnegative, $H(X \mid Y) \geq 0$, it follows that

$$H(Y) \leq H(X),$$

with equality if and only if $H(X \mid Y) = 0$, which holds if and only if X is determined by Y—that is, if T is injective on the support of $p(x)$. $\square$

Remark 3.3. The quantity $H(X \mid Y) = H(X) - H(Y)$ measures the information about the source that is lost in the normalization process. It represents, in bits, the diversity of generative inputs that cannot be recovered from the normalized output. When this quantity is large, the normalization layer has compressed substantial generative diversity into a uniform representation—but the source diversity nonetheless exists and is not destroyed by the transformation.

3.4 The Data Processing Inequality

The proposition above is a special case of the more general data processing inequality, which states that no processing of a signal can increase the information it carries about its source [Shannon, 1948]. If $X \to Y \to Z$ is a Markov chain, then

$$I(X; Z) \leq I(X; Y),$$

where $I(\cdot; \cdot)$ denotes mutual information. Applied to our setting, this means that any further transformation of the normalized output Y—including reception and interpretation by a human reader—can only further reduce the information available about the generative source X.

The data processing inequality formalizes a core asymmetry in the communication system: information flows from source to receiver but cannot be increased along the way. Uniformity in Y is therefore not merely compatible with diversity in X; it is what one should *expect* from a many-to-one normalization layer applied to a diverse source. Observing that $H(Y)$ is small provides no evidence about the magnitude of $H(X)$.

3.5 Formal Implications for the Homogenization Thesis

The implications for the homogenization thesis are direct. If LLMs operate primarily as normalization transformations applied to human-generated inputs, then the observation that LLM-assisted texts exhibit reduced stylistic variability is precisely what the channel model predicts—regardless of whether the generative source has changed at all. The reduced entropy of Y is a property of the transformation T, not of the source distribution $p(x)$.

Demonstrating cognitive homogenization requires evidence that $H(X)$ has declined, not merely that $H(Y)$ is low. These are logically independent claims. Any study that measures only properties of the outputs Y is measuring the wrong quantity.

4 Private Generative Systems and Public Registers

The formal argument of the preceding section acquires additional force when situated within the broader history of normalization in human communication systems. The coexistence of private generative diversity and public representational uniformity is not a novel feature introduced by AI systems. It is a structural characteristic of literate societies that has been reproduced across multiple historical periods and communicative technologies.

4.1 Historical Normalization Mechanisms

Spelling standardization in the early modern period dramatically reduced the orthographic diversity observable in texts without homogenizing the ideas those texts expressed. Before the consolidation of standard orthographies in the seventeenth and eighteenth centuries, written texts exhibited extraordinary variation in spelling, punctuation, and typographic practice—variation that reflected genuine diversity in regional pronunciation, scribal convention, and individual preference. The introduction of standardized dictionaries and printing practices collapsed this variation into a small set of conventional forms. Yet no historian of ideas has argued that orthographic standardization produced intellectual homogenization.

Editorial institutions in scientific publishing impose strong and consistent expectations on rhetorical structure, citation practice, and argumentative organization. A paper submitted to a journal in any major scientific discipline will pass through a normalization process that aligns its surface form with disciplinary conventions. The diversity of ideas published across such journals vastly exceeds the diversity of their rhetorical presentation. The normalization of form has not constrained the diversity of content.

Translation systems present a particularly instructive analogy. When a text is translated from one language to another, the translation necessarily loses certain features of the original—phonological texture, idiomatic resonance, lexical ambiguity—while preserving, imperfectly but substantially, the conceptual content. The translated text is a compressed projection of the original. Multiple distinct originals might produce highly similar translations. This compression does not imply that the original texts were cognitively equivalent.

4.2 LLMs as Automated Normalization Layers

Large language models can be understood as extending this historical lineage of normalization mechanisms into a new technological register. When used as editing or rewriting tools, they perform a function structurally similar to that of a human editor, style guide, or translator: they map the author's idiosyncratic expression into a form that conforms more closely to widely shared conventions and can therefore be more easily interpreted by a broader audience.

The novelty of LLMs relative to prior normalization mechanisms lies principally in three dimensions: their computational scale, their adaptability across domains and registers, and their accessibility to users who previously lacked access to professional editorial services. None of these dimensions alters the basic logical structure of the normalization relationship between source and channel.

4.3 The Expansion of Expressive Space

A counterintuitive possibility emerges from this analysis. If large language models effectively function as translation layers between private generative systems and shared public registers, their availability may reduce the pressure that communicating agents experience to conform to normative conventions during the act of composition itself.

Prior to the availability of automated normalization, a writer who preferred an unconventional spelling system, a non-prestige dialect, or a multilingual mixing of registers faced a direct trade-off: communicate idiosyncratically and sacrifice audience, or conform to conventions and sacrifice expressive authenticity. Automated normalization dissolves this trade-off. The writer can compose in whatever register is most generatively natural and allow the normalization layer to perform the translation when communication is required.

Under these conditions, the availability of automated normalization may expand rather than contract the diversity of private linguistic practice. The apparent uniformity of the final output would reflect the existence of a translation interface, not the impoverishment of the generative source.

This possibility is not merely speculative. There is historical precedent in the introduction of word processing software, which reduced the mechanical cost of revision and thereby enabled a more exploratory approach to composition. Whether LLMs produce analogous effects on expressive diversity is an empirical question, but the theoretical possibility is well-grounded.

5 Instrumental and Epistemic Interaction

The preceding analysis focused on cases where LLMs are used instrumentally as editing or translation tools. In such cases, the human author retains primary control over the conceptual content of the communication and uses the model merely to normalize its surface form. However, LLMs are also used in modes that involve a more substantive epistemic relationship between the user and the model's outputs. Distinguishing these modes of interaction is essential for a precise assessment of the homogenization thesis.

5.1 Instrumental Use

In instrumental use, the model functions as a tool for drafting, translation, summarization, or formatting. The user specifies the conceptual content—the ideas to be expressed, the arguments to be made, the information to be conveyed—and recruits the model to assist with the linguistic form of the expression. The model's contribution is primarily syntactic and stylistic rather than conceptual.

Under instrumental use, the source-channel model developed above applies with maximal force. The human author's generative processes operate largely independently of the model's distribution, and the model's outputs carry minimal information about those processes beyond what the author's inputs already contain. Any convergence in the final texts reflects the normalization performed by the channel, not the convergence of the underlying sources.

5.2 Epistemic Use

In epistemic use, the user approaches the model as a source of factual information, analysis, or interpretive guidance. The user's initial conceptual framework is incomplete or uncertain, and model-generated outputs contribute to its formation. In this mode, the model functions as a cognitive authority rather than merely a linguistic tool.

Epistemic use raises more substantive concerns about perspectival convergence. If large numbers of users consult the same systems for information and analysis, and if those systems produce consistent outputs that reflect the distributional biases of their training data, then users' epistemic frameworks may be systematically influenced in converging directions.

However, several qualifications limit the force of this concern. First, even in epistemic use, model outputs are typically not passively accepted but are evaluated against prior beliefs, independent evidence, and alternative sources. The degree of epistemic influence depends critically on the user's prior state of knowledge and critical orientation. Second, demonstrating lasting epistemic influence requires longitudinal evidence distinguishing temporary exposure effects from durable changes in belief or reasoning strategy—evidence that existing studies have not yet provided.

5.3 The Importance of the Distinction

The distinction between instrumental and epistemic use matters because the homogenization thesis tends to elide it. Studies that observe stylistic convergence in model-assisted texts are studying a phenomenon that occurs even in purely instrumental use, and therefore carry no implications for epistemic homogenization. Studies that observe temporary opinion alignment following exposure to opinionated model outputs are studying a phenomenon that occurs even in cases where the influence does not persist—and that therefore carry no implications for durable cognitive convergence.

A rigorous assessment of the homogenization hypothesis requires empirical designs that are sensitive to the distinction between these modes and that measure the appropriate dependent variables for each.

6 Linguistic Attractor Basins

The homogenization thesis implicitly assumes that large language models impose a single dominant attractor within the space of possible linguistic expressions: a single peak or mode in the output distribution toward which all model-assisted texts converge. This assumption is empirically false in a straightforward sense—large language models produce diverse outputs across a wide range of registers, styles, and subject matters—but the underlying structural concern deserves more precise analysis.

Definition 6.1 (Linguistic Space)**.** Let $\mathcal{L}$ denote the space of possible linguistic expressions in a given language or family of languages. Points in $\mathcal{L}$ represent individual utterances, texts, or discourse units, endowed with whatever geometric or topological structure is appropriate to the analysis.

Definition 6.2 (Model Distribution)**.** Let $P_\theta : \mathcal{L} \to [0, 1]$ denote the probability distribution over expressions induced by a language model with parameters θ. The model assigns higher probability to expressions that appear more frequently in configurations similar to those seen during training.

Definition 6.3 (Linguistic Attractor Basin)**.** A *linguistic attractor basin* of a language model with distribution P_θ is a region $A \subset \mathcal{L}$ such that expressions within A have relatively high probability under P_θ and perturbations of those expressions remain within A after transformation by the normalization layer.

Several features of this definition deserve elaboration. First, attractor basins need not be unique. A language model trained on diverse stylistic corpora will develop multiple attractor regions corresponding to different registers, genres, rhetorical forms, and subject domains. The output distribution is multimodal, not concentrated at a single point.

Second, the existence of attractor basins in the model's distribution does not imply that human generative processes collapse into those basins. When humans use language models instrumentally, their generative processes may explore large and diverse regions of $\mathcal{L}$ before the normalization layer projects the resulting expressions into nearby high-probability regions. The projection reduces entropy at the output without constraining the range of inputs.

Third, the attractor basins of a model evolve over time as the model's parameters and fine-tuning objectives change. There is no fixed homogenizing target; the normalization surface itself is dynamic.

Proposition 6.4 (Non-Injectivity and Basin Structure)**.** *Suppose $A \subset \mathcal{L}$ is an attractor basin of P_θ and let $U \supset A$ be a neighborhood of A in $\mathcal{L}$. If the normalization transformation T maps all points in U to points in A, then T restricted to U is non-injective for any $|U| > |A|$, and consequently*

$$H(T(X)) < H(X)$$

for any source distribution $p(x)$ supported on U.

Proof. If $|U| > |A|$ and T maps U into A, then by the pigeonhole principle, at least two distinct points in U share the same image in A under T. The transformation is therefore non-injective, and the entropy inequality $H(T(X)) < H(X)$ follows from Proposition 3.2. □

7 The Conceptual Landscape

The attractor structure of a language model can be visualized geometrically by interpreting the logarithm of the model probability as a potential landscape. Define the potential function $V : \mathcal{L} \to \mathbb{R}$ by

$$V(x) = -\log P_\theta(x).$$

Regions of high probability correspond to valleys (local minima of V), while low-probability regions correspond to elevated terrain. The attractor basins identified in the previous section correspond geometrically to the valleys of this potential landscape.

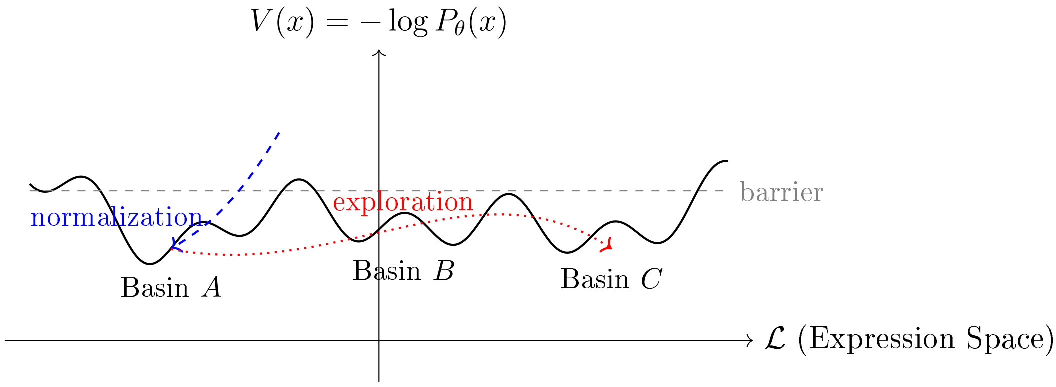

Figure 1: Potential landscape $V(x) = -\log P_\theta(x)$ showing three attractor basins A, B, C. Normalization (blue dashed) projects expressions from surrounding terrain into nearby basins. Human exploratory generation (red dotted) may traverse the landscape widely before normalization applies.

Several features of this landscape are significant for the present argument. First, models trained on diverse corpora develop landscapes with multiple valleys rather than a single global minimum. The normalized output may cluster within any of several basins rather than collapsing to a single point.

Second, human linguistic creativity characteristically involves the exploration of varied terrain: combining structures from different valleys, ascending from familiar basins to discover neighboring regions, or venturing into rarely visited high-probability subspaces that prior work has not mapped. The normalization layer applies after this exploration, projecting the resulting expression into the nearest basin without erasing the exploratory process that generated it.

Third, the landscape itself is a function of the model parameters θ, which evolve across model versions and fine-tuning objectives. The targets of normalization shift over time; there is no fixed "average" toward which all expression is inexorably drawn.

7.1 Attractor Dynamics Under Repeated Interaction

A more nuanced concern arises when users interact repeatedly with language models in a mode that is neither purely instrumental nor purely epistemic but somewhere in between. In such interactions, the model's outputs may gradually influence the forms of expression that feel natural to the user—not by constraining the content of thought, but by providing a familiar stylistic template that the user unconsciously reproduces.

This effect is real and has plausible historical analogues. Writers who read extensively in a particular genre often find that its stylistic conventions infiltrate their own prose. Researchers trained in a specific disciplinary tradition tend to adopt its characteristic rhetorical forms. Whether repeated interaction with model-generated text produces comparable effects is an empirical question, but the mechanism is not implausible.

The important point, however, is that such stylistic influence does not constitute cognitive homogenization in the sense relevant to the thesis. The homogenization thesis is concerned with diversity in ideas, perspectives, and reasoning strategies—the diversity of the generative source in the conceptual sense. Stylistic normalization, even if it reaches into the composition process rather than applying only as a post-hoc transformation, is compatible with the full persistence of conceptual diversity.

8 Exploratory Entropy and the Ecology of Knowledge

The preceding analysis has argued that generative diversity at the level of the cognitive source is unlikely to be substantially reduced by channel normalization. This does not, however, exhaust the culturally relevant consequences of LLM-mediated communication. A distinct and more subtle form of homogenization may operate not at the level of expression but at the level of *discovery*: the pathways through which individuals encounter ideas, encounter conflicting interpretations, and develop the conditions for genuine conceptual innovation.

8.1 Exploratory Entropy Defined

Definition 8.1 (Knowledge Environment)**.** Let $\mathcal{S} = \{s_1, s_2, \ldots, s_n\}$ denote the set of informational sources available within a knowledge environment. These include books, articles, archival materials, personal correspondence, course lectures, and any other medium through which ideas are encountered. Let $P(s_i)$ denote the probability that a learner encounters source s_i during an episode of intellectual exploration.

Definition 8.2 (Exploratory Entropy)**.** The *exploratory entropy* of a knowledge environment is

$$H_E = -\sum_{s \in \mathcal{S}} P(s) \log P(s).$$

High exploratory entropy corresponds to a discovery process in which exposure is distributed broadly across the available sources; low exploratory entropy corresponds to a process in which a small subset of sources dominates exposure, regardless of the size of $\mathcal{S}$.

Remark 8.3. Exploratory entropy is distinct from both source entropy $H(X)$ and channel entropy $H(Y)$. It concerns the diversity of informational trajectories through which ideas are encountered, rather than the diversity of expressions produced or received. A system can maintain high $H(X)$ and high $H(Y)$ while exhibiting low H_E—that is, individuals may continue to produce and receive diverse expressions while all encountering the same small set of ideas along the same conceptual pathways.

8.2 Traditional Knowledge Environments

Traditional knowledge environments—physical libraries, disciplinary reading lists, archival research, informal scholarly correspondence—characteristically maintained relatively high exploratory entropy. The distribution $P(s)$ was shaped by the intersection of multiple independent factors: geographic proximity to particular collections, disciplinary training, personal recommendations, and the productive chance of finding unexpected materials in the course of searching for something else. These factors created overlapping but partially independent pathways through the available corpus, exposing different individuals to different subsets of available ideas and thereby generating the conditions for diverse intellectual trajectories.

High exploratory entropy also enables what might be called *cross-basin discovery*: the encounter with ideas that inhabit a different region of the conceptual landscape from the region one's prior work occupies. Such encounters are a significant source of intellectual innovation, since they introduce unexpected analogies, challenge presupposed frameworks, and open new angles of approach to persistent problems.

8.3 Algorithmic Mediation and the Concentration of Discovery

Generative AI systems, by contrast, typically synthesize information from many sources into a single summarized representation. The user who queries a language model for an explanation of a complex topic receives a single canonicalized output that collapses the diversity of sources and interpretations into one streamlined narrative.

Formally, let $\mathcal{C}$ be a corpus of documents. A generative summary $\mathcal{G}(\mathcal{C})$ is a compression operator

$$\mathcal{C} \to \mathcal{G}(\mathcal{C})$$

that maps the full corpus onto a single synthesized representation. Users who rely on $\mathcal{G}(\mathcal{C})$ rather than directly consulting the corpus encounter a highly concentrated distribution over ideas: the ideas that the synthesis surface prominently, weighted by their frequency and the model's training distribution.

If a large proportion of users rely on the same generative systems to navigate the same knowledge domains, the distribution $P(s)$ over sources becomes increasingly concentrated around the canonical outputs of those systems. Exploratory entropy declines—not because the corpus has shrunk or because individuals have become less capable of diversity in their own expression, but because the pathways through which they encounter the corpus have contracted.

Proposition 8.4 (Entropy Concentration Under Summarization)**.** *Let $\mathcal{C} = \{c_1, \ldots, c_m\}$ be a corpus and let $P_0(c_i) = 1/m$ be the uniform discovery distribution. Suppose a generative summary assigns saliency weights $w_i \geq 0$ to each document, normalized so that $\sum_i w_i = 1$. Define the summary-mediated discovery distribution as $P_G(c_i) = w_i$. If the weights are nonuniform—that is, if $\max_i w_i > 1/m$—then*

$$H_E(P_G) < H_E(P_0) = \log m.$$

Proof. Since entropy is uniquely maximized by the uniform distribution over a finite set, any nonuniform distribution P_G over $\{c_1, \ldots, c_m\}$ satisfies $H_E(P_G) < \log m = H_E(P_0)$. □

8.4 The Significance of Exploratory Variance

The cultural significance of exploratory entropy lies in its relationship to intellectual innovation and adaptive resilience. From the perspective of cultural evolution, the resilience of a knowledge ecosystem depends not only on the diversity of ideas it currently contains but on the probability that novel combinations and reframings will be generated in the future. This probability is higher when exploratory pathways are diverse, because diversity in discovery creates the conditions for unexpected cross-fertilization.

If the mechanisms of discovery increasingly converge toward centralized systems that surface the same canonical framings to large populations of users, the expected rate of conceptual recombination may decline. Many users may encounter similar interpretive schemas even if their individual expressive styles remain entirely distinct. The cultural consequence would not be a reduction in stylistic variation but a narrowing of the conceptual routes through which ideas are encountered—a form of homogenization that operates at the level of the epistemic infrastructure rather than at the level of individual expression.

9 Rank-Mediated Discovery and Its Entropy

Modern information environments rarely expose users directly to the full set of available sources. Instead, search engines, recommender systems, and generative models mediate access through ranking, filtering, and summarization mechanisms. Each of these mechanisms systematically transforms the discovery distribution in ways that reduce exploratory entropy.

9.1 Ranking Functions and Exposure Distributions

Let $\mathcal{S} = \{s_1, \ldots, s_n\}$ be a set of available sources. A ranking function R assigns to each source an ordinal rank $R(s_i) \in \{1, 2, \ldots, n\}$. A monotonically decreasing exposure function $f : \{1, \ldots, n\} \to (0, 1]$ translates ranks into exposure probabilities. The rank-mediated discovery distribution is

$$P'(s_i) = \frac{f(R(s_i))}{\sum_{j=1}^{n} f(R(s_j))}.$$

In practice, f decays sharply with rank position. Studies of user behavior on search engines consistently find that the probability of clicking on a result falls off rapidly beyond the first few ranked items [Lazer et al., 2009, Raghavan and Hanna, 2023]. A first-ranked source may receive orders of magnitude more exposure than a tenth-ranked source.

Proposition 9.1 (Ranking Reduces Exploratory Entropy)**.** *Let P_0 be the uniform distribution over $\mathcal{S}$. For any monotonically decreasing exposure function f that is non-constant,*

$$H_E(P') < H_E(P_0) = \log n.$$

Proof. The rank-mediated distribution P' is nonuniform whenever f is non-constant, since sources at different rank positions receive different exposure probabilities. By the uniqueness of the maximum-entropy distribution over a finite set, $H_E(P') < \log n = H_E(P_0)$. □

9.2 Compounding Effects of Ranking and Summarization

Generative AI systems introduce an additional layer of compression beyond ranking. Rather than merely ordering sources, they synthesize multiple ranked sources into a single output. If the user receives only the synthesized output rather than the ranked list, the effective discovery distribution collapses from P' to a point mass on the synthesized summary—an entropy of zero.

In practice, users retain some residual exposure to original sources through citations and links, and some proportion of users actively pursue original materials.

The effective discovery distribution is a mixture between the point-mass on the summary and the original distribution P' over cited sources. But even in the most optimistic case, the entropy of this mixture is bounded above by the entropy of P', which is itself bounded by $\log n$. Summarization introduces a floor on entropy reduction that stacks multiplicatively with the reduction introduced by ranking.

9.3 Design as a Variable

It is important to note that the entropy of the rank-mediated discovery distribution depends on the specific design of the information interface, not merely on the existence of algorithmic mediation. Interfaces that surface multiple alternative summaries, provide explicit citation links to diverse original sources, incorporate stochastic exploration mechanisms, or present conflicting interpretations alongside consensus views will maintain higher exploratory entropy than interfaces that present a single canonical response.

The policy implications are therefore not that generative AI systems necessarily reduce exploratory entropy, but that their design choices determine by how much they reduce it. This is a tractable engineering and institutional problem, not a structural inevitability.

10 Normalization Layers in Historical Communication Systems

The analysis developed in this paper situates large language models within a long historical tradition of normalization mechanisms in communication systems. This situating serves a dual function: it both demystifies the LLM case by showing that its structural logic is not novel, and it furnishes historical evidence relevant to the empirical question of whether normalization mechanisms have historically produced cognitive convergence.

10.1 Case Studies in Communicative Standardization

Three historical cases merit detailed consideration.

Standardized orthography. The emergence of standardized spelling in European languages between the sixteenth and eighteenth centuries compressed an extraordinary range of orthographic variation into a small set of conventional forms. The linguistic diversity of the pre-standardization period—in which educated writers might spell the same word differently on the same page—is well documented by historians of language. Its compression did not visibly impoverish the diversity of ideas expressed in the resulting standardized texts. The century following orthographic standardization in England was the century of the Scientific Revolution. If anything, standardization facilitated the cross-regional communication of diverse ideas by reducing the cognitive cost of decoding nonstandard orthographic forms.

Scientific editorial conventions. The development of standardized scientific prose in the late seventeenth and eighteenth centuries—associated with institutions such as the Royal Society and the Académie des sciences—imposed strong rhetorical norms on scientific writing. The passive voice, the impersonal register, the systematic use of tables and mathematical notation, and the conventional structure of hypothesis, method, result, and discussion are all artifacts of this normalization process. None of these conventions has been shown to have constrained the diversity of scientific ideas expressed within them; on the contrary, they facilitated the communication of diverse findings across linguistic and disciplinary barriers.

Translation. The practice of translation has always involved the compression of generative diversity. A translation system—whether human or automated—maps the rich generative space of one language onto the different generative space of another, inevitably losing features of the original in the process. Yet translation has historically expanded the effective audience of ideas rather than homogenizing them, by making diverse ideas accessible across language barriers that would otherwise prevent their communication.

10.2 The Structural Pattern

These historical cases share a common pattern. In each, a normalization mechanism applied to the communication channel reduced the surface diversity of the observable output without reducing the diversity of the ideas that output expressed. In some cases, normalization actively facilitated the dissemination of diverse ideas by reducing the cognitive barriers to comprehension across heterogeneous audiences.

The same structural pattern applies to LLM-mediated normalization. The novel features of LLMs—their computational power, their breadth across domains, their accessibility—may amplify both the normalization effects and the facilitating effects. Which of these amplifications dominates is a contingent empirical question, not one that can be settled by theoretical analysis alone.

11 Reexamining Recent Empirical Claims

Recent discussions of linguistic and cognitive homogenization have been articulated most explicitly in the work of Sourati et al. [2026], who argue that large language models may gradually standardize human expression, perspective, and reasoning. Their analysis raises important concerns regarding the cultural consequences of generative AI systems and has stimulated a broader scholarly conversation about the dynamics of human–machine communication.

The information-theoretic and ecological framework developed in the preceding sections provides several grounds for reconsidering specific assumptions that underlie their claims.

11.1 Representation Versus Generation

A central premise of the homogenization argument is that similarities in linguistic output reflect convergence in the underlying cognitive processes that produced those outputs. As the formal analysis of Section 3 demonstrates, this premise is false as a general logical claim. A normalization layer can produce output similarity from diverse inputs; observing output similarity is therefore not evidence of input similarity.

For the premise to hold, it would need to be established that the normalization layer in question is approximately injective—that different cognitive processes tend to produce distinctly different model-assisted outputs. This claim requires positive evidence, not merely the assumption that output tracks input.

11.2 Instrumental Versus Epistemic Interaction

The homogenization argument often treats model outputs as epistemic influences that substantively reshape users' conceptual frameworks. The distinction developed in Section 5 shows that this characterization applies only to epistemic modes of interaction. In instrumental modes, the model functions as a communication interface, and its influence on underlying reasoning is limited.

Existing empirical studies of homogenization have not, in general, adequately controlled for the difference between these modes. Studies that expose participants to opinionated model-generated content are studying the epistemic interaction mode, and their results cannot be generalized to instrumental use cases.

11.3 Short-Term Effects and Long-Term Dynamics

The empirical studies cited in support of homogenization typically employ short-term experimental designs: participants interact with model outputs during a single session, and subsequent shifts in expressed opinion or linguistic style are measured. Such designs are capable of detecting immediate alignment effects but provide no evidence about whether those effects persist once the interaction ends.

The relevant claim in the homogenization thesis is about long-term cultural dynamics: a gradual and durable shift in the generative processes of a population. Demonstrating such a shift requires longitudinal designs with appropriate washout periods and measures that can distinguish persistent changes in generative behavior from temporary surface alignment. The existing literature provides no such evidence.

11.4 Stylistic Similarity and Conceptual Diversity

Many studies supporting the homogenization thesis focus on stylistic or lexical features of text: the distribution of word frequencies, syntactic constructions, or rhetorical organizational patterns. These are precisely the features that normalization layers are designed to standardize. As noted above, observing stylistic convergence in model-assisted texts is consistent with—and indeed predicted by—a channel normalization account.

The relevant measure for the homogenization thesis is not stylistic similarity but *conceptual diversity*: the diversity of ideas, arguments, and interpretive frameworks expressed in the texts. Measuring conceptual diversity is substantially more difficult than measuring stylistic similarity, but it is the measure that the thesis actually requires.

12 Toward a Precise Research Agenda

The preceding critiques identify specific methodological gaps in the existing empirical literature. Filling these gaps requires a research agenda organized around a clearer conceptual framework. We propose three priority directions.

12.1 Longitudinal Studies of Generative Behavior

The central empirical question is whether sustained use of LLMs produces durable changes in the generative processes of human authors—changes that persist in the absence of model assistance. Addressing this question requires longitudinal study designs that track the same individuals over extended periods, with periodic measurement of unassisted writing under controlled conditions.

Such studies should measure not only surface stylistic features but also conceptual properties: the range of arguments made, the diversity of sources cited, the variety of framing strategies employed. If generative diversity contracts over time in unassisted writing, and if this contraction correlates with the intensity of model use, the homogenization thesis would receive genuine empirical support.

12.2 Transfer Studies in Reasoning

The reasoning convergence hypothesis requires evidence of durable transfer from model-mediated reasoning contexts to independent problem-solving behavior. Experimental designs could expose participants to extended interaction with model-generated reasoning traces for specific problem types and then measure problem-solving strategies on novel instances of the same types in the absence of model assistance. If transfer occurs and if the transferred strategies resemble model-generated strategies more than alternative human strategies for the same problems, this would constitute evidence for reasoning convergence.

12.3 Measurement of Exploratory Entropy

The concept of exploratory entropy introduced in Section 8 suggests a third research direction: direct measurement of how algorithmic mediation reshapes the distribution of sources that individuals encounter. Digital trace data from browsing behavior, citation networks, and search interaction logs could in principle be used to estimate $P(s)$ before and after the introduction of model-mediated search into a given domain.

Comparing the entropy of discovery distributions across populations with varying degrees of model-mediated versus direct-search engagement would provide direct evidence about the ecological effects of generative AI on knowledge exploration.

13 Cognitive Prostheses and Human Agency

Generative AI systems are most productively understood within the broader theoretical tradition of cognitive prostheses: technologies that externalize aspects of cognitive processing and thereby reconfigure the functional boundaries of the human cognitive system.

13.1 The Prosthetic Tradition

Writing itself is the paradigm case of a cognitive prosthesis. By externalizing memory into inscribed symbols, writing extended the temporal and spatial reach of human cognition in ways that fundamentally altered the structure of intellectual culture. Yet no one concludes from the universality of written language that human cognition has homogenized; rather, writing created new conditions for the proliferation of diverse ideas by enabling their stable transmission across time and space.

Printing extended this effect by distributing written content to larger audiences at lower cost. Calculators externalized arithmetic processing, freeing cognitive resources for higher-order mathematical reasoning. Search engines externalized the process of bibliographic exploration, enabling access to larger corpora than any individual could maintain in memory.

Each of these prosthetic technologies reshaped the landscape of cognitive activity without eliminating human agency in directing that activity. They introduced new affordances and new constraints, shifted the allocation of cognitive effort, and created new dependencies. But the direction of inquiry, the selection of questions, and the evaluation of conclusions remained irreducibly human.

13.2 LLMs as Prosthetic Systems

Large language models extend this lineage into new domains. They externalize substantial portions of the linguistic production process: drafting, editing, reformulation, summarization. They also externalize portions of the information retrieval and synthesis process: answering questions, explaining concepts, generating literature reviews.

The empirically relevant question is not whether this externalization influences human cognitive behavior—it surely does, as do all prosthetic technologies—but how it reshapes the space of available cognitive strategies. The history of prior prostheses suggests that the answer to this question is rarely simple and often counterintuitive. Calculators did not reduce mathematical thinking; they shifted it from arithmetic to conceptual domains. Search engines did not reduce reading; they changed the patterns of reading. The effects of LLMs on human cognition are likely to exhibit comparable complexity.

13.3 Human Agency in the Prosthetic System

A central feature of prosthetic technologies is that they do not merely constrain the human who uses them but create new possibilities for action. The writer using a word processor has access to revision strategies that would be prohibitively costly with pen and paper. The researcher using a search engine can explore connections across bodies of literature that would be inaccessible within a physical library's limits.

LLMs similarly create new possibilities: the ability to quickly prototype arguments, to translate idiomatic expression into standard forms, to generate alternative framings of a familiar problem. Whether individuals exploit these possibilities in ways that expand or contract the diversity of their intellectual output is a function of how they choose to use the technology—a function, in other words, of human agency operating within the new landscape the technology creates.

This observation has normative implications for the homogenization debate. If the cultural consequences of LLMs depend substantially on how they are used, the appropriate response to concerns about homogenization is not merely to analyze the statistical properties of model outputs but to design usage contexts, interface features, and pedagogical practices that encourage expansive rather than contractive modes of engagement.

14 Implications for Linguistic Diversity

The theoretical framework developed throughout this paper permits a more nuanced and precise account of LLMs' effects on linguistic diversity than the homogenization thesis provides. Several implications warrant explicit statement.

14.1 Normalization and Diversity Are Compatible

The central formal result is that normalization in the communication channel is compatible with the full preservation of generative diversity at the source. Proposition 3.2 establishes this compatibility in the general case, and the historical cases of Section 10 provide empirical corroboration. There is no logical pathway from the observation of normalized outputs to the conclusion of cognitive homogenization.

14.2 Normalization May Expand Effective Expressive Space

The argument of Section 4 suggests that automated normalization may expand rather than contract the effective expressive space of individual authors by removing the communicative penalty that previously attached to non-standard expression. This possibility is underexplored in existing literature and merits empirical investigation.

14.3 The Ecological Risk Is Real but Differently Located

The framework identifies a genuine cultural risk associated with LLM-mediated communication, but locates it differently from the homogenization thesis. The risk lies not in the normalization of expressive form but in the potential contraction of exploratory entropy: the diversity of pathways through which individuals encounter ideas. Propositions 8.4 and 9.1 formalize the conditions under which this contraction occurs and demonstrate that it is a structural feature of summarization and ranking systems rather than a contingent effect.

14.4 Design Choices Determine Cultural Consequences

The magnitude of the exploratory entropy reduction introduced by AI systems depends substantially on their design: the specificity of their summaries, the diversity of sources they surface, the degree to which they support rather than replace direct engagement with original materials, and the stochasticity of their discovery mechanisms. These are tractable design choices, not structural inevitabilities.

15 Conclusion: Diversity, Compression, and Cultural Dynamics

The homogenization thesis rests on the assumption that uniformity in linguistic outputs implies convergence in human cognition. This assumption is formally invalid: a normalization layer in the communication channel can produce the former without the latter. The data processing inequality—that post-processing cannot increase mutual information—establishes this result as a general theoretical claim, not merely a possibility in specific cases.

Large language models, when used instrumentally as editing, translation, or drafting tools, operate as normalization layers. They map diverse generative inputs onto a more standardized output distribution. The entropy of that output is necessarily lower than the entropy of the source distribution; but this reduction is a property of the transformation, not evidence that the source distribution has changed. Observing stylistic convergence in LLM-assisted texts no more establishes cognitive convergence than observing orthographic uniformity in printed texts establishes uniformity in the ideas those texts express.

The genuine cultural risk identified by this analysis lies elsewhere. If the mechanisms of knowledge discovery become increasingly centralized and algorithmically mediated, exploratory entropy may decline even while expressive diversity persists. The diversity of conceptual pathways through which individuals encounter ideas—a significant source of intellectual innovation and cultural resilience—may contract as a consequence of the summarization and ranking dynamics that characterize current AI systems.

This is a tractable policy and design problem. Interfaces that preserve exploratory diversity, that surface conflicting interpretations alongside canonical ones, that encourage direct engagement with original sources, and that expose users to unexpected conceptual neighborhoods can maintain higher exploratory entropy than those that optimize exclusively for efficiency and convenience. The cultural consequences of generative AI depend on these choices in ways that are amenable to deliberate intervention.

Understanding the long-term implications of generative language technologies therefore requires sustained attention to both dimensions of the system: the diversity of generative sources and the structure of the channels through which ideas circulate. These are distinct variables, governed by distinct mechanisms, and they require distinct empirical and normative frameworks. Conflating them—as the homogenization thesis characteristically does—produces misdirected concern and obscures the places where cultural vigilance is most needed.

Appendices

A Formal Derivation of Entropy Inequalities Under Deterministic Normalization

This appendix collects and extends the entropy-theoretic results discussed in the main text.

A.1 Setup

Let $(\mathcal{X}, p)$ be a discrete probability space representing the generative source, where $\mathcal{X}$ is the set of possible linguistic expressions and $p(x) = \mathbb{P}(X = x)$ is the source distribution. Let $\mathcal{Y}$ be the output space and $T : \mathcal{X} \to \mathcal{Y}$ a deterministic function representing the normalization transformation. The observable output is the random variable $Y = T(X)$, with distribution

$$p(y) = \mathbb{P}(Y = y) = \mathbb{P}(T(X) = y) = \sum_{\{x:T(x)=y\}} p(x).$$

A.2 Entropy of the Source

$$H(X) = -\sum_{x\in\mathcal{X}} p(x) \log p(x).$$

A.3 Entropy of the Normalized Output

$$H(Y) = -\sum_{y\in\mathcal{Y}} p(y) \log p(y) = -\sum_{y\in\mathcal{Y}} \left(\sum_{\{x:T(x)=y\}} p(x) \right) \log \left(\sum_{\{x:T(x)=y\}} p(x) \right).$$

A.4 The Core Inequality

Theorem A.1 (Entropy Reduction Under Deterministic Normalization)**.** *Let X be a discrete random variable and $Y = T(X)$ a deterministic function of X. Then*

$$H(Y) \leq H(X),$$

with equality if and only if T is injective on the support of $p(x)$.

Proof. Since Y is a deterministic function of X, the conditional entropy satisfies $H(Y \mid X) = 0$. By the chain rule:

$$H(X, Y) = H(X) + H(Y \mid X) = H(X).$$

Also by the chain rule:

$$H(X, Y) = H(Y) + H(X \mid Y).$$

Therefore $H(X) = H(Y) + H(X \mid Y)$, so $H(Y) = H(X) - H(X \mid Y) \leq H(X)$ since $H(X \mid Y) \geq 0$.

For equality: $H(X \mid Y) = 0$ if and only if X is a function of Y, which holds if and only if T is injective on the support of $p(x)$. $\square$

A.5 Quantifying Information Loss

The information lost in normalization is measured by the equivocation $H(X \mid Y)$:

$$H(X \mid Y) = H(X) - H(Y) = \sum_{y \in \mathcal{Y}} p(y) H(X \mid Y = y),$$

where

$$H(X \mid Y = y) = -\sum_{\{x : T(x) = y\}} \frac{p(x)}{p(y)} \log \frac{p(x)}{p(y)}.$$

This conditional entropy measures the residual diversity within each equivalence class $T^{-1}(y)$. When normalization collapses many diverse expressions into a single output, $H(X \mid Y = y)$ is large for that y, reflecting the substantial generative diversity that is invisible in the normalized representation.

A.6 The Data Processing Inequality

The main theorem is a consequence of the more general data processing inequality. If Z is any random variable such that $X \to Y \to Z$ is a Markov chain (i.e., Z depends on X only through Y), then

$$I(X; Z) \leq I(X; Y),$$

where $I(X; Y) = H(X) - H(X \mid Y)$ denotes mutual information. For deterministic $Y = T(X)$, $I(X; Y) = H(Y)$, and the inequality reduces to our main result.

B Exploratory Entropy: Formal Treatment

B.1 Setup

Let $\mathcal{S} = \{s_1, \ldots, s_n\}$ be a finite set of informational sources and P a probability distribution over $\mathcal{S}$ representing the discovery probability. The exploratory entropy is

$$H_E(P) = -\sum_{i=1}^{n} P(s_i) \log P(s_i).$$

B.2 Maximum Entropy and the Uniform Distribution

By the maximum-entropy principle, $H_E(P)$ is uniquely maximized by the uniform distribution $P_0(s_i) = 1/n$, at which point $H_E(P_0) = \log n$.

B.3 Entropy Under Ranking

Let $R : \mathcal{S} \to \{1, \ldots, n\}$ be a bijective ranking function and $f : \{1, \ldots, n\} \to (0, \infty)$ a strictly decreasing exposure function. Define the rank-weighted distribution:

$$P'(s_i) = \frac{f(R(s_i))}{\sum_{j=1}^{n} f(R(s_j))}.$$

Since f is strictly decreasing, P' is nonuniform, and therefore $H_E(P') < \log n = H_E(P_0)$.

Lemma B.1 (Rank Concentration)**.** *For any strictly decreasing f, the rank-mediated distribution P' satisfies*

$$H_E(P') < H_E(P_0).$$

Moreover, as f becomes more steeply decreasing (in the sense that the ratio $f(1)/f(n)$ increases), $H_E(P')$ decreases monotonically toward zero.

Proof. The first claim follows from the strict concavity of the entropy function on the probability simplex: the unique maximizer of H_E is P_0, and $P' \neq P_0$ for any nonconstant f. The second claim follows from the observation that as $f(1)/f(n) \to \infty$, all probability mass concentrates on the top-ranked source, and $H_E(P') \to 0$. $\square$

B.4 Entropy Under Summarization

Let $G : \mathcal{P}(\mathcal{S}) \to \mathcal{S}_*$ be a generative summarization operator that maps subsets of $\mathcal{S}$ to a summary space $\mathcal{S}_*$. If $\mathcal{S}_* = \{*\}$ (a single canonical summary), the discovery distribution degenerates to a point mass with $H_E = 0$.

In the realistic mixed case where users access the summary with probability α and access original sources (distributed according to P') with probability $1 - \alpha$, the effective discovery distribution is the mixture

$$P_\alpha = \alpha \cdot \delta_* + (1 - \alpha) \cdot P',$$

where δ_* is the point mass on the summary. The entropy of this mixture satisfies

$$H_E(P_\alpha) \leq (1-\alpha)H_E(P') + h(\alpha),$$

where $h(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha)$ is the binary entropy function. This bound shows that increasing α (greater reliance on AI summaries) monotonically reduces the upper bound on exploratory entropy.

C A Unified Proposition on Normalization and Exploration

Proposition C.1 (Dual Independence of Normalization and Exploratory Entropy). *The following hold simultaneously:*

(i) Normalization layers may reduce observable linguistic entropy $H(Y)$ while leaving generative cognitive entropy $H(X)$ unchanged.

(ii) Algorithmic discovery systems may reduce exploratory entropy H_E even when both $H(X)$ and $H(Y)$ remain unchanged.

(iii) Consequently, $H(X)$, $H(Y)$, and H_E are mutually logically independent in AI-mediated communication systems.

Proof. Claims (i) and (ii) follow from Theorem A.1 and Lemma B.1 respectively. For claim (iii), independence follows from the construction of explicit examples:

High $H(X)$, low $H(Y)$, low H_E: Diverse authors ($H(X)$ high) all use the same AI editor ($H(Y)$ low) and all consult the same AI-mediated discovery channel (H_E low).

High $H(X)$, low $H(Y)$, high H_E: Diverse authors ($H(X)$ high) all use the same AI editor ($H(Y)$ low) but explore independently through diverse libraries (H_E high).

High $H(X)$, high $H(Y)$, low H_E: Diverse authors with minimal AI editing ($H(Y) \approx H(X)$ high) but who all rely on the same AI-curated news feed (H_E low). □

Remark C.2. This proposition clarifies the structure of the homogenization debate. The three quantities $H(X)$, $H(Y)$, and H_E capture three distinct dimensions of diversity in an AI-mediated knowledge system: the diversity of generative cognitive processes, the diversity of observable linguistic outputs, and the diversity of epistemic discovery pathways. Each is a meaningful and distinct object of study. Conflating them—as the homogenization thesis tends to do—produces a misdirected research program that measures the wrong variables and draws unwarranted conclusions.

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